Sujal Bangde

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SUMMARY

Software Developer with experience in Python, Django, FastAPI, Node.js, React.js, and SQL databases. Proficient in building full-stack applications, RESTful APIs, secure authentication systems, and scalable backend solutions. Skilled in Generative AI, LLMs, LangChain, and RAG-based applications, with a strong interest in developing intelligent and modern software solutions.

EDUCATION

S B Jain Institute of Technology Management & Research **Nagpur**
Bachelor of Technology in Computer Science & Engineering Aug. 2020 - May 2024

Lokmanya Tilak Mahavidyalaya **Wani, Yavatmal**
Higher Secondary Certificate February-2020

EXPERIENCE

Full Stack Developer **January 2025 – Present**
Sagveek Technologies Pvt Ltd Pune

- Develop and maintain full-stack web applications using Python, FastAPI, Django, Node.js, Express.js, React.js, Next.js, and MySQL/PostgreSQL.
- Integrate backend systems with frontend applications and ensuring high-performance API consumption.
- Integrate third-party services, including email services, authentication systems, and external APIs to enhance application functionality.
- Implement secure JWT-based authentication and role-based access control mechanisms.
- Design and optimizing relational databases, write efficient SQL queries, and managing data migration processes.
- Implement file upload, export, reporting, and automated email notification functionalities.
- Designed and developed an offline multilingual Retrieval-Augmented Generation (RAG) system for intelligent search and analysis of Marathi, Hindi, and English land-record documents.
- Built document ingestion pipelines using PyMuPDF, PDFPlumber, Tesseract OCR, OpenCV, and Poppler to process native, scanned, and degraded PDFs.
- Implemented hybrid retrieval mechanisms using multilingual embeddings, BM25 search, Reciprocal Rank Fusion, and reranking techniques to improve answer relevance.
- Integrated LangChain, Qdrant, Ollama, and Qwen 2.5 to generate context-aware and source-grounded responses through locally hosted LLMs.
- Developed FastAPI-based services for document upload, ingestion monitoring, querying, health checks, and streaming response generation.

INTERNSHIPS

Python Developer **January 2024 – August 2024**
Sagveek Technologies Pvt Ltd Pune

1) Hospital Management System.

- Engineered a role-based Hospital Management System with separate Admin, Doctor, and Patient modules. Using Python, Django, REST APIs, PostgreSQL, HTML, CSS, Bootstrap.
- Designed and implemented RESTful APIs for seamless frontend-backend communication and data integration.
- Worked on database design, optimization, and management using PostgreSQL.
- Executed secure user authentication and authorization mechanisms for different user roles.
- Built CRUD operations for doctors, patients, appointments, and medical records.
- Implemented doctor availability management and appointment conflict prevention.
- Sent automated email reminders before scheduled appointments.
- Generated appointment status notifications (Pending, Approved, Rejected).
- Built appointment scheduling and patient record management functionalities.

2) Organization Contact Management System.

- Architected a centralized contact management platform for storing and managing organizational contacts. Using Python, Django, REST APIs, PostgreSQL, HTML, CSS, BootStrap.
- Applied advanced search and filtering features for quick contact access.
- Created responsive and user-friendly interfaces using Bootstrap and modern web technologies.
- Established secure authentication and role-based access controls.
- Designed and implemented RESTful APIs for contact creation, retrieval, updating, and deletion.

Backend Developer **July 2023 - August 2023**
Whitehorse Business Solution Pvt Ltd Nagpur

- Developed a Flask web application for Uploading, Extracting, and Automating the testing of ZIP files.
- Executed Computerized ZIP file validation and secure storage mechanisms, reducing invalid file uploads and improving data integrity.

Intern Web Developer **August 2022 - September 2022**
Tech Vegan LTD Nagpur

- Utilized Mobirise to design responsive web pages and Figma to create UI/UX prototypes, enhancing user interface design and improving the user experience.
- Developed responsive websites using Mobirise based on client requirements and business needs.

PROJECTS

Multilingual RAG System for Land-Record Intelligence | Python, FastAPI March 2025 – January 2026

- Built an offline multilingual RAG pipeline designed to process up to 400,000 pages of land records, legal acts, mutation entries, and property documents.
- Implemented page-level PDF triage using PyMu PDF, PDFPlumber, Tesseract OCR, OpenCV, and Poppler to handle native, scanned, degraded, and mixed-layout PDFs.
- Combined dense vector search with BM25 sparse retrieval, Reciprocal Rank Fusion, and cross-encoder reranking to improve retrieval relevance.
- Generated multilingual vector embeddings with multilingual-e5-large and stored them in Qdrant for semantic retrieval.
- Added text normalization for Devanagari OCR errors and legacy Marathi/Hindi fonts such as Shivaji and Kruti Dev.
- Developed resumable ingestion, SHA-256 deduplication, confidence-based OCR review, metadata filtering, and JSONL audit logging.
- Created a FastAPI backend with document upload, health monitoring, ingestion status, querying, and streaming-response endpoints.
- Provided command-line tools using Typer for PDF ingestion, individual and batch queries, BM25 rebuilding, and OCR reprocessing.

Virtual Snake Game (Computer Vision) | Python, CV Zone, OpenCV September 2023 - January 2024

- Built Snake game controlled by hand gestures, leveraging computer vision to capture hand movements via a webcam.
- Integrated Open CV for real-time video processing and hand gesture recognition to control the direction of the snake.
- Utilized CV Zone for efficient gesture tracking and mapping gestures to snake movement, enhancing user interaction.

Travel Advisor Website | HTML, CSS, Node.js, JavaScript December 2022 - February 2023

- Developed Travel Advisor website to offer personalized travel recommendations, helping users plan their trips.
- Planned the front-end with HTML and CSS for a responsive interface and integrated Node.js for server-side operations, handling user requests and data processing.
- Integrated Map APIs to provide real-time navigation, location-based recommendations, and interactive maps for travel exploration.

Talking Chat Bot | Python, Tkinter, Chatter Bot May 2022 - June 2022

- Created an interactive Chat Bot using Python, Tkinter for the user interface, and Chatter Bot for generating conversational responses.
- Enabled both text and voice input for user interaction, allowing users to ask questions and receive responses based on pre-set answers.

Digital Clock | C June 2021 - June 2021

- Streamlined a digital clock using the C programming language. The application displays the time in a digital format, providing a user-friendly interface for timekeeping.

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, JavaScript.

AI & Gen AI: Generative AI, LLMs, LangChain, RAG.

Frontend: HTML5, CSS3, Bootstrap, React.js, Next.js, Material UI.

Backend: Django, Django REST Framework, FastAPI, RESTful APIs, Flask, Node.js, Express.js.

ML & NLP: PyTorch, Sentence Transformers, Embeddings, OCR, Tesseract OCR.

Databases: PostgreSQL, MySQL, MongoDB, SQLite, Qdrant.

Tools: VS Code, Visual Studio, PyCharm, Git, GitHub, Postman.